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1. Introduction

METALOGOS is an AI-native programming language built on seven pillars: Entity, Pattern, Flow, Memory, Rule, Learn, and Adapt. The language is implemented in Rust and follows a contract-first TDD approach with golden tests. Phase 3 represents the final stage of the developer tooling milestone, completing the transition from a research prototype to a language ecosystem suitable for real-world development workflows.

Phase 3 delivered four major components: (1) a CLI with subcommands for running, checking, and interactively exploring .mlog programs; (2) a standard library with import mechanism providing string, math, and collections modules; (3) an LSP server for editor integration with real-time diagnostics, go-to-definition, hover, and completion; and (4) a package manager for project initialization, dependency tracking, and build validation. Additionally, comprehensive mdbook documentation was created covering tutorials, syntax reference, standard library reference, and an index of all Architecture Decision Records.

All 44 tests across three workspace crates pass successfully. The entire Phase 3 has been committed and pushed to the GitHub repository ShkodnikAI/Metalogos-. This report provides a detailed overview of every sub-phase, the architectural decisions made, the test strategy, and the final project structure.

2. Project Overview

The METALOGOS workspace consists of three Cargo crates organized as a Cargo workspace, alongside standard library modules, documentation, and editor support files. The workspace was progressively expanded throughout Phase 3, starting with two members (core + CLI binary) and ending with three dedicated crates plus the main binary.

2.1 Workspace Structure

Component	Path	Purpose
metalogos (lib)	src/	Core library: parser, AST, interpreter, semantic analysis, LLM client, ML backend, embedding backend
mlog (binary)	src/main.rs	CLI: mlog run repl check
mlog-lsp	mlog-lsp/	LSP server: diagnostics, go-to-definition, hover, completion (tower-lsp)
mlogpkg	mlogpkg/	Package manager: init add build info
Standard Library	std/	std/string.mlog, std/math.mlog, std/collections.mlog
Examples	examples/	12 .mlog programs with .expected golden outputs
VS Code Extension	editors/vscode/	package.json manifest + TextMate grammar + language config

Documentation	docs/book/	mdbook: tutorial, syntax reference, stdlib reference, ADR index
ADRs	docs/adr/	12 Architecture Decision Records (0001-0019)

Table 1: Workspace components

2.2 Metrics

Metric	Value
Total Rust source lines	4,530
Core library (src/)	3,266 lines across 8 files
LSP server (mlog-lsp/)	487 lines
Package manager (mlogpkg/)	348 lines
Grammar (grammar.pest)	155 rules
Integration tests	140 lines across 4 files
Golden examples	12 .mlog + .expected pairs
Standard library modules	3 (string, math, collections)
mdbook chapters	4 (tutorial, syntax, stdlib, ADR index)
ADRs written	12 total (3 in Phase 3: 0017, 0018, 0019)
Total test count	44 passed, 0 failed

Table 2: Project metrics

3. Phase 3.1: CLI + REPL + Semantic Check

Phase 3.1 transformed the single-command binary into a full CLI with three subcommands and introduced an interactive REPL with persistent state. Prior to this phase, the only way to execute a .mlog program was via a single ``mlog run`` command. The interpreter was stateless, creating a fresh instance on every invocation and discarding all state after completion. This made incremental development and interactive exploration impossible.

3.1.1 CLI Architecture

The CLI uses clap derive macros with three subcommands. The ``mlog run`` subcommand executes a .mlog program, loading the standard library from the base directory of the source file for proper import resolution. The ``mlog check`` subcommand performs semantic analysis without execution, returning errors and warnings with exit code 0 for clean programs and exit code 1 for errors. The ``mlog repl``

subcommand starts an interactive session with line editing, command history (persisted to `~/.mlog_history`), and a persistent Interpreter instance across all inputs.

The REPL has two modes: interactive (tty) mode using rustyline with a "mlog>" prompt and command history, and piped (non-tty) mode that reads lines from stdin silently for use in integration tests. TTY detection uses `libc::isatty(0)` on Unix systems, with a fallback `METALOGOS_FORCE_PIPE=1` environment variable for test control. A new `feed_line()` function in `lib.rs` parses a single line into declarations and feeds them to the existing interpreter, allowing entities, patterns, memory, relations, and adapted patterns to survive between inputs.

3.1.2 Semantic Analysis

A new semantic module (`src/semantic.rs`) provides `check_program()` which performs two-pass analysis on declarations without executing them. The first pass collects all declaration names and detects duplicates. The second pass validates cross-references: entity types referenced in records exist, field initializers reference valid fields, patterns and learnable patterns invoked in flows exist, flow branch targets are known patterns, rule targets reference existing entities, and adapt/mutate targets reference existing learnable patterns. The result is an `AnalysisResult` containing separate errors and warnings vectors, each with a `format()` method for display.

3.1.3 Tests

Test	File	Validates
repl_integration_three_lines	tests/repl_integration.rs	entity + pattern + flow incremental eval via pipe
check_ok_program	tests/check_integration.rs	valid program produces no errors
check_undefined_type_error	tests/check_integration.rs	reports unknown type error
check_adapt_target_not_found	tests/check_integration.rs	reports missing learnable pattern
check_duplicate_entity_type	tests/check_integration.rs	reports duplicate entity type
check_format_no_issues	tests/check_integration.rs	"OK: no issues found" for clean program
5 semantic unit tests	src/semantic.rs	ok program, undefined type, adapt target, duplicate, detailed spans

Table 3: Phase 3.1 test suite (11 tests)

4. Phase 3.2: Standard Library + Import

Phase 3.2 introduced the standard library and the import mechanism, enabling code reuse across `.mlog` programs. The standard library provides three modules with commonly needed patterns that wrap built-in functions (prefixed with double underscore to avoid name collisions with user-defined patterns).

4.1 Import Mechanism

The import statement has the form ``import std/module``, which triggers file path resolution at runtime. The interpreter maintains a base directory (either the current working directory for REPL or the parent directory of the source file for ``mlog run``) and resolves imports relative to a `std/` subdirectory. The imported file is parsed and its declarations are merged into the current interpreter state, making all patterns from the imported module available for use in the importing program. Duplicate declarations across modules are handled by the existing duplicate detection in semantic analysis.

4.2 Standard Library Modules

Module	Patterns	Description
std/string	trim, replace, split, join	String manipulation: whitespace trimming, substring replacement, splitting by separator, joining lists with separator
std/math	abs, min, max, clamp, round	Mathematical operations: absolute value, minimum/maximum of two values, range clamping, rounding to nearest integer
std/collections	first, last, push	Basic collection operations: get first or last element, append item to list

Table 4: Standard library modules

Each standard library pattern is a thin wrapper around a built-in function. Built-in functions use double underscore prefixes (e.g., `__trim`, `__abs`, `__first`) to avoid conflicts with user-defined patterns. The standard library patterns provide clean, human-readable names that users import explicitly.

5. Phase 3.3: LSP Server

Phase 3.3 delivered a full Language Server Protocol implementation using the `tower-lsp` framework, providing real-time editor integration for `.mlog` files. The LSP server runs as a separate binary (`mlog-lsp`) communicating via `stdio`, following the standard LSP transport protocol. This was the first workspace expansion, adding `mlog-lsp` as a Cargo workspace member.

5.1 Span Propagation

LSP features require source positions for diagnostics, go-to-definition, and hover. The existing AST carried no position data, so a `Span` struct was added with `start_line`, `start_col`, `end_line`, and `end_col` fields (0-indexed, LSP-compatible). Every declaration struct received a `span` field with Default trait implementation for backward compatibility. The parser was modified with a `span_from_pair()` helper that extracts positions from `pest Pair::line_col()` at each declaration boundary. Helper methods were added to `Declaration`: `span()`, `name()` for symbol table lookup, `kind_str()` for hover info, and `type_info()`

for type signature display.

5.2 Spanned Diagnostics

A new `AnalysisResultDetailed` type was added alongside the existing `AnalysisResult` in `semantic.rs`. It includes a `Vec` of `Diagnostic` objects with `span`, `message`, and `severity` fields. The `check_program_detailed()` function produces these spanned diagnostics, while `check_program()` continues to return the simple format for CLI backward compatibility. The LSP server converts these diagnostics to LSP `Diagnostic` objects using `span_to_range()` mapping.

5.3 LSP Features

Feature	Implementation
Diagnostics	<code>textDocument/didOpen</code> and <code>textDocument/didChange</code> trigger parse + semantic analysis + publish diagnostics
Go-to-definition	Builds symbol table from declarations; finds identifier at cursor via text scanning; matches to declaration span
Hover	Finds declaration at cursor; returns type signature and kind as Markdown
Completion	Lists all named declarations from the current document
Text sync	FULL mode (full text on every change)

Table 5: LSP features implemented

5.4 VS Code Extension

A VS Code extension manifest (`editors/vscode/package.json`) declares the `.mlog` language with file extension association, a `TextMate` grammar for syntax highlighting, and configuration options for the LSP server path and trace level. The extension activates on `.mlog` files and launches the `mlog-lsp` binary as an LSP client.

5.5 LSP Tests

Test	Contract
<code>lsp_did_open_erroneous_program_returns_diagnostic</code>	Erroneous program triggers ERROR severity diagnostic with "unknown type" message and valid range
<code>lsp_did_open_clean_program_no_diagnostics</code>	Clean program produces zero diagnostics
<code>lsp_did_open_parse_error_returns_diagnostic</code>	Invalid syntax triggers parse error diagnostic
<code>lsp_did_open_multiple_errors_multiple_diagnostics</code>	Multiple errors produce multiple diagnostics

lsp_did_open_duplicate_pattern_returns_error	Duplicate pattern detected
lsp_did_open_adapt_not_found_returns_error	Missing adapt target detected
lsp_goto_definition_finds_entity	Entity declaration found via symbol table lookup
lsp_hover_returns_type_info	Pattern type_info contains signature with parameter types
10 unit tests	Word extraction, span conversion, symbol building, multi-line, bounds checking

Table 6: LSP test suite (18 tests)

6. Phase 3.4: Package Manager (mlogpkg)

The final Phase 3 deliverable is mlogpkg, a package manager for .mlog projects. It provides project initialization with a standard manifest format, dependency tracking, build validation, and lock files for reproducible builds. This was the second workspace expansion, adding mlogpkg as the third Cargo workspace member.

6.1 Manifest Format

The mlog.toml manifest uses a TOML format familiar to Rust developers, mirroring the Cargo.toml structure. It contains a [package] section with name, version, and edition fields, plus a [dependencies] section mapping package names to version constraints. The default edition is "2024" for forward compatibility with future language editions.

```
[package]
name = "my-project"
version = "0.1.0"
edition = "2024"

[dependencies]
some-pkg = "0.3.0"
```

6.2 Commands

Command	Description
mlogpkg init [--name N]	Creates mlog.toml manifest and src/main.mlog scaffold. Name defaults to directory name.
mlogpkg add <pkg> [--version V]	Adds a dependency to mlog.toml. Verifies package exists in local registry (~/.mlog/registry/).
mlogpkg build	Resolves dependencies from local registry, collects all .mlog files from src/, runs semantic check on each, writes mlog.lock.

mlogpkg info	Displays project name, version, edition, dependencies, entry point, and registry path.
--------------	--

Table 7: mlogpkg commands

6.3 Local Registry

Packages are stored in `~/.mlog/registry/<pkg-name>/`. Each package directory contains its own `mlog.toml` manifest and source files. There is no remote server at this phase; packages are installed manually by copying files into the registry directory. This intentionally minimal approach provides a foundation for future remote registry support with `publish/fetch` commands, semantic version constraint resolution, and package signing.

6.4 Build Workflow

The `mlogpkg build` command executes a seven-step pipeline: (1) read `mlog.toml`, (2) find entry point (`src/main.mlog` by default), (3) resolve dependencies from local registry, (4) collect all `.mlog` source files recursively from `src/`, (5) run semantic analysis via `metalogs::check_program` on each file, (6) write `mlog.lock` with resolved versions, and (7) report success with file count or failure with error count. The lock file uses JSON format and records each dependency's resolved version and source origin.

6.5 Package Manager Tests

Test	Validates
<code>test_init_creates_mlog_toml</code>	init creates <code>mlog.toml</code> with name and version fields + <code>src/main.mlog</code> scaffold
<code>test_init_default_name</code>	init without <code>--name</code> uses directory name
<code>test_init_fails_if_exists</code>	init fails when <code>mlog.toml</code> already exists
<code>test_build_with_init_project</code>	build succeeds on fresh project, creates <code>mlog.lock</code>
<code>test_build_detects_errors</code>	build fails when source has semantic errors
<code>test_build_fails_without_manifest</code>	build fails without <code>mlog.toml</code>
<code>test_info_no_manifest</code>	info gracefully reports no <code>mlog.toml</code>
<code>test_info_with_manifest</code>	info shows project name and version
<code>test_manifest_toml_format</code>	manifest has <code>[package]</code> , <code>[dependencies]</code> , edition fields

Table 8: mlogpkg test suite (9 tests)

7. Documentation (mdbook)

Comprehensive mdbook documentation was created in docs/book/ with four chapters covering the full language from beginner to advanced usage. The documentation serves as both a learning resource and a reference guide for the METALOGOS language.

7.1 Tutorial

The tutorial (docs/book/src/tutorial.md) is a 10-step hands-on guide that progressively introduces all language features. It starts with a minimal "Hello, World!" program using entity, pattern, and flow, then builds up through entity types and records, learnable patterns (LLM-powered), adaptation with adapt and mutate, sandbox for safety, fluid types with confidence scores, rules for conditional logic, memory constructs, standard library imports, and finally a complete pipeline combining all features. Each step includes executable code examples with expected output.

7.2 Syntax Reference

The syntax reference (docs/book/src/syntax.md) provides a complete specification of all surface syntax constructs: entity (simple, type, record), pattern, learnable pattern, flow (linear and branching), fluid, rule, memory (memorize, forget), adaptation (adapt, mutate, sandbox), import, and relate. It also documents all expression types (string literal, float literal, identifier, field access, function call, binary operators), built-in functions, comparison operators, and file conventions. The reference serves as the authoritative source for syntax questions.

7.3 Standard Library Reference

The standard library reference (docs/book/src/stdlib.md) documents all three standard library modules with function signatures, descriptions, and usage examples for each pattern. It covers std/string (trim, replace, split, join), std/math (abs, min, max, clamp, round), and std/collections (first, last, push).

7.4 ADR Index

The ADR index (docs/book/src/adr-index.md) lists all 12 Architecture Decision Records from 0001 through 0019, providing a navigable reference to the project's architectural history and design rationale.

8. Architecture Decision Records

Three ADRs were written during Phase 3, documenting the key architectural decisions for each sub-phase. Each ADR follows the standard format: context, decision, and consequences (positive, negative, future).

8.1 ADR-0016: CLI + REPL + Semantic Check

This ADR documents the decision to use clap derive macros for the CLI, rustyline for the REPL (chosen over reedline for lower complexity), and a two-pass semantic analysis approach. The REPL achieves state persistence by keeping a single Interpreter instance alive across iterations via the new feed_line() public API. The semantic check module validates programs without execution, enabling CI integration and quick feedback without running potentially slow LLM calls. Key trade-off: REPL state is in-memory only (not persisted between sessions), and semantic check is structural only without full type inference.

8.2 ADR-0018: LSP Server

This ADR documents the decision to create a separate mlog-lsp crate using tower-lsp (the de-facto standard LSP framework for Rust with async support via tokio and full LSP 3.16 compatibility). Spans were added to all declaration AST nodes for source position tracking. Go-to-definition and hover use a text-scanning approach rather than adding spans to every expression node, which would require pervasive AST changes. Text sync uses FULL mode, acceptable for .mlog files which are typically under 1000 lines. Known limitation: pest's line_col() may report slightly different line numbers for programs with blank lines.

8.3 ADR-0019: Package Manager

This ADR documents the decision to create a separate mlogpkg crate as a workspace member, using a TOML manifest format mirroring Cargo.toml for familiarity. The local registry approach (~/.mlog/registry/) was chosen as intentionally minimal for Phase 3, deferring remote registry support. The build workflow validates all source files with the existing semantic analysis module. Limitations: no version constraint resolution (uses exact versions), no workspace/multi-package support, and manual package installation required.

9. Test Summary

The complete test suite across all three workspace crates consists of 44 tests, all passing. The tests cover unit-level logic, golden-file acceptance testing, CLI integration via subprocess, REPL integration via stdin piping, LSP integration via core code path testing, and package manager integration via filesystem-based subprocess testing.

Crate	Test File	Count
metalogos	src/semantic.rs (unit)	5
metalogos	tests/check_integration.rs	5
metalogos	tests/golden.rs	1 (auto-discovers 12 pairs)
metalogos	tests/repl_integration.rs	1

mlog-lsp	src/lib.rs (unit)	10
mlog-lsp	tests/lsp_integration.rs	8
mlogpkg	tests/pkg_integration.rs	9

Table 9: Complete test suite (44 tests)

10. Git History

Phase 3 produced four commits on the main branch, each corresponding to a sub-phase:

Commit	Message
747500c	Phase 3: CLI + REPL + semantic check (ADR-0016)
e9ce815	Phase 3.2: Standard Library (import, std/string, std/math, std/collections)
ae32174	Phase 3.3: LSP server (mlog-lsp on tower-lsp)
a9d5891	Phase 3 closed: package manager (mlogpkg), mdbook docs, ADR 0019

Table 10: Phase 3 Git commits

11. Conclusion and Next Steps

Phase 3 is complete. The METALOGOS language now has a comprehensive developer tooling ecosystem: CLI with run/repl/check subcommands, interactive REPL with persistent state and history, semantic analysis for CI integration, standard library with import mechanism, LSP server for real-time editor support (diagnostics, go-to-definition, hover, completion), package manager for project management, and thorough documentation covering tutorials, syntax reference, standard library reference, and ADR index.

The language successfully demonstrates all seven pillars (Entity, Pattern, Flow, Memory, Rule, Learn, Adapt) with 12 example programs, 44 passing tests, and 12 ADRs documenting architectural decisions. The workspace structure supports clean separation of concerns with three independent crates sharing the core library.

Phase 4 is the next milestone: bytecode VM/JIT compilation and self-hosting. This will involve designing a bytecode instruction set, building a stack-based virtual machine, implementing a JIT compiler for hot paths, and ultimately compiling the METALOGOS interpreter to run on its own bytecode format, achieving self-hosting capability.